1. Project Title: Autonomous Noise and Air Quality Monitoring Robot

2. Purpose

Objective :

To design and build an autonomous robot that can move around a defined area while measuring air quality and noise levels, avoiding obstacles, and displaying or sending the collected data for environmental analysis.

Scope:

This robot can be used in urban areas, classrooms, factories, or campuses to monitor pollution levels. It will navigate autonomously, detect air and noise conditions, and alert users if levels exceed safe limits. The robot's data can help identify pollution hotspots and support environmental awareness initiatives.

Significance:

Air and noise pollution are serious environmental and health threats. Manual data collection is time-consuming and limited. This project promotes smart environmental monitoring, low-cost automation, and supports sustainable smart city development using basic robotics and sensor technology.

3. Components

Category	Component Name	Details / Purpose
Microcontroller	Arduino Uno R3	Controls all sensors and actuators
Sensors	MQ135 Gas Sensor	Measures air quality (CO_2, smoke, ammonia, etc.)

	Sound Sensor Module	Measures environmental noise in decibels
	Ultrasonic Sensor (HC-SR04)	Detects obstacles for navigation
Actuators	DC Motors (2 units)	Enables movement of the robot
	Servo Motor	Rotates sensor head or triggers alerts
Body/Chassis	2-Wheel Robot Chassis Kit	Base frame with motor mounts and caster wheel
Additional Components	L298N Motor Driver, LCD Display, Battery Pack, Breadboard, Jumper Wires, LEDs	For motor control, power supply, and data display

4. Cost Breakdown (Approx.)

Components	Unit Cost	Total

5. Functionality Breakdown

Functionality 1: Air Quality Monitoring

Overview:

The robot measures the surrounding air pollution using an MQ135 gas sensor.

Working Procedure:

- The MQ135 sensor continuously detects CO_2, NH_3, and smoke particles.

- The sensor output is read by the Arduino’s analog input pin.
- Data is displayed on an LCD screen or sent via Bluetooth.
- If air quality crosses a threshold, an LED or buzzer alert is activated

Functionality 2: Noise Level Monitoring

Overview:

Measures noise intensity using a sound sensor module.

Working Procedure:

- The sound sensor detects variations in environmental noise.
- The Arduino processes the analog signal and converts it to decibel values.
- If noise exceeds safe levels, an LED/buzzer alert triggers automatically.

Functionality 3: Autonomous Navigation and Data Collection

Overview:

The robot moves around to scan multiple areas autonomously while avoiding obstacles.

Working Procedure:

- The ultrasonic sensor detects obstacles and sends distance data to the Arduino.
- The Arduino adjusts motor directions using the L298N driver to avoid collisions.
- Data from all sensors (noise + air quality) is collected during movement.
- The robot can stop for short intervals to take accurate readings before moving again.

Feedback from teacher we got:

- [optional] Add zig-zag or grid-scan navigation so the robot collects data from multiple locations instead of moving randomly.
- MQ135 must be calibrated; use fresh-air baseline and define numeric thresholds for pollution levels.
- Convert the sound sensor analog value into approximate dB levels or set threshold ranges for safe/moderate/loud noise.
- Include LED or buzzer alerts for unsafe air or noise readings to improve demonstration clarity.
- Physically isolate the sound sensor from motors to avoid vibration- related false noise reading.